

Commonly available power ratings are 0.4, 1, 3, 10, and 50 watts.

Example 4.1. Determine the temperature coefficient of a 5 V zener diode at 25°C, if the nominal voltage drops to 4.8 V at a temperature of 100°C.

Solution: Change in zener voltage, $\Delta V_Z = 4.8 - 5 = -0.2$ V
Change in temperature, $(t - t_0) = 100 - 25 = 75^\circ\text{C}$.

$$\text{Temperature coefficient of zener diode} = \frac{\Delta V_Z}{V_Z (t - t_0)} \times 100 = \frac{-0.2}{5 \times 75} \times 100 = -0.0533\% \text{ Ans.}$$

Example 4.2. A 15 V zener diode is connected in series with a forward-biased silicon diode for constructing a zero-temperature coefficient voltage reference. The temperature coefficient of the silicon diode is $-1.7 \text{ mV}/^\circ\text{C}$. Find the required temperature coefficient of the zener diode in per cent per degree.

[U.P.S.C. I.E.S. Electronics & Telecommunication Engineering-I, 2012]

Solution: To compensate the zener diode temperature effect both the temperature coefficient should be the same, so that we will get zero-temperature coefficient voltage reference

$$\text{So } \Delta V_Z/^\circ\text{C} = 1.7 \text{ mV}/^\circ\text{C}$$

Required temperature coefficient of zener diode in per cent per degree

$$= \frac{\Delta V_Z}{V_Z} \times 100\% \text{ per degree} = \frac{1.7 \times 10^{-3}}{15 \text{ V}} \times 100 = 0.01133\%/^\circ\text{C} \text{ Ans.}$$

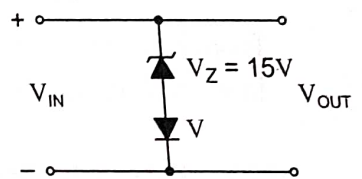


Fig. 4.5

Example 4.3. For the circuit shown in Fig. 4.7, find

- (a) the output voltage, (b) the voltage drop across R_S ,
(c) the current through zener.

Solution: From the figure shown in Fig. 4.8 we have

(a) Output voltage, $V_L = V_Z = 50 \text{ V}$ Ans.

(b) Voltage drop across $R_S = I_S \times R_S = V_S - V_Z = 120 - 50 = 70 \text{ V}$ Ans.

(c) Load current, $I_L = \frac{V_L}{R_L} = \frac{50 \text{ V}}{10 \text{ k}\Omega} = 5 \text{ mA}$

Current through R_S , $I_S = \frac{70 \text{ V}}{5 \text{ k}\Omega} = 14 \text{ mA}$

Hence, current through zener diode is given by

$$I_Z = I_S - I_L = 14 \text{ mA} - 5 \text{ mA} = 9 \text{ mA} \quad \text{Ans.}$$

Example 4.4. For the voltage circuit in Fig. 4.9, determine the maximum and minimum I_Z .

[Uttarakhand Technical Univ. Fundamentals of Electronics Engineering Odd Semester, 2009-10]

Solution:

$$\text{Load current, } I_L = \frac{V_{\text{OUT}}}{R_L} = \frac{V_Z}{R_L} = \frac{50}{10 \text{ k}\Omega} = 5 \text{ mA}$$

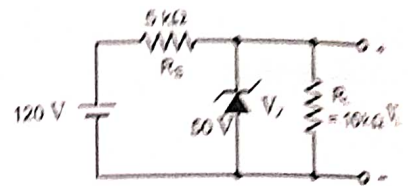


Fig. 4.7

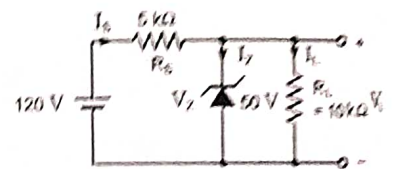


Fig. 4.8

Special Diodes

Zener current, I_Z will be maximum when input voltage is maximum i.e. 200 V.

Corresponding current through series resistance, R_S of 5 k Ω ,

$$I_{S \max} = \frac{V_{S \max} - V_Z}{R_S} = \frac{200 - 50}{5 \text{ k}\Omega} = 30 \text{ mA}$$

Corresponding zener current, $I_{Z \max} = I_{S \max} - I_L = 30 - 5 = 25 \text{ mA}$ Ans.

Zener current, I_Z will be minimum when input voltage is minimum i.e. 100 V.

Corresponding current through series resistance, R_S of 5 k Ω ,

$$I_{S \min} = \frac{100 - 50}{5 \text{ k}\Omega} = 10 \text{ mA}$$

Corresponding zener current, $I_{Z \min} = I_{S \min} - I_L = 10 - 5 \text{ mA}$ Ans.

Example 4.5. For the zener diode network of Fig. 4.10, determine V_L , V_R , I_Z and P_Z .

[Mahamaya Technical Univ. Electronics Engg. Second Semester 2010-11]

Solution: Voltage across load if the zener diode is open circuited

$$V_L = \frac{V_{in} \times R_L}{R + R_L} = \frac{16 \times 1.2}{1 + 1.2} = 8.727 \text{ V}$$

Since voltage across load is less than the breakdown voltage of zener diode i.e., $V_L < V_Z$, the zener diode will not conduct and voltage across load, $V_L = 8.727 \text{ V}$ Ans.

Voltage across series resistor, $V_R = 16 - 8.727 = 7.273 \text{ V}$ Ans.

Zener diode current, $I_Z = 0$ Ans.

Zener diode power, $P_Z = V_Z \times I_Z = V_Z \times 0 = 0$ watt Ans.

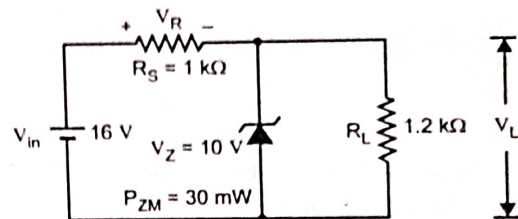


Fig. 4.10

$\therefore$ Zener diode is open circuited

Example 4.6. For the circuit shown in Fig. 4.11, $R_S = 20 \Omega$, $V_Z = 18 \text{ V}$ and $R_L = 200 \Omega$. If V_{in} can vary from 20 V to 30 V, find

- The minimum and maximum currents in zener diode.
- The minimum and maximum power dissipated in the diode.
- The maximum rated power dissipation that R_S should have.

[G.G.S.I.P. Univ. Analog Electronics, December-2009]

Solution:

$$\text{Load current, } I_L = \frac{V_{out}}{R_L} = \frac{V_Z}{R_L} = \frac{18}{200} = 0.09 \text{ A}$$

$$\text{Maximum supply current, } I_{S \max} = \frac{V_{in \max} - V_Z}{R_S} = \frac{30 - 18}{20} = 0.6 \text{ A}$$

$$(i) \text{ Corresponding zener current, } I_{Z \max} = I_{S \max} - I_L = 0.6 - 0.09 = 0.51 \text{ A Ans.}$$

$$\text{Minimum supply current, } I_{S \min} = \frac{V_{in \min} - V_Z}{R_S} = \frac{20 - 18}{20} = 0.1 \text{ A}$$

$$\text{Corresponding zener current, } I_{Z \min} = I_{S \min} - I_L = 0.1 - 0.09 = 0.01 \text{ A or } 10 \text{ mA Ans.}$$

$$(ii) \text{ Minimum power dissipation in diode } = V_Z \times I_{Z \min} = 18 \times 0.01 = 0.18 \text{ W Ans.}$$

$$\text{Maximum power dissipation in diode } = V_Z \times I_{Z \max} = 18 \times 0.51 = 9.18 \text{ W Ans.}$$

$$(iii) \text{ Maximum power dissipation rating of } R_S$$

$$= I_{S \max}^2 R_S = 0.6^2 \times 20 = 7.2 \text{ W Ans.}$$

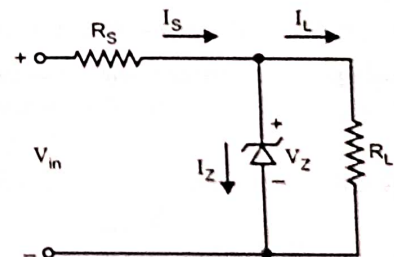


Fig. 4.11

Example 4.7. For the network of Fig. 4.12, determine the range of R_L and I_L that will result in V_{RL} being maintained at 10 V. Also determine the maximum wattage rating of diode.

[G.B. Technical Univ. Electronics Engineering First Semester 2011-12]

Solution:

$$\begin{aligned}\text{Input voltage, } V_{in} &= 50 \text{ V} \\ \text{Series resistance, } R &= 1 \text{ k}\Omega \\ V_Z &= 10 \text{ V} \\ I_{Zm} &= 32 \text{ mA}\end{aligned}$$

$$\text{Supply current, } I_R = \frac{V_{in} - V_L}{R} = \frac{50 - 10}{1,000} = 40 \text{ mA}$$

Load current I_L will be maximum when zener current

$$I_Z = 0$$

$$\text{So } I_{L \max} = I_R - I_{Z \min} = 40 - 0 = 40 \text{ mA}$$

Corresponding load resistance will be minimum and

$$R_{L \min} = \frac{V_L}{I_{L \max}} = \frac{V_Z}{I_{L \max}} = \frac{10}{40 \times 10^{-3}} = 250 \Omega$$

Load current I_L will be minimum when I_Z is maximum i.e., 32 mA.

$$\text{So } I_{L \min} = I_R - I_{Z \max} = 40 - 32 = 8 \text{ mA}$$

Corresponding load resistance will be maximum and

$$R_{L \max} = \frac{V_L}{I_{L \min}} = \frac{10}{8 \times 10^{-3}} = 1.25 \text{ k}\Omega$$

Hence range of R_L : From 250 Ω to 1.25 k Ω Ans. and range of I_L : From 8 mA to 40 mA Ans.

$$\text{Maximum wattage rating of diode} = V_Z \times I_{Z \max} = 10 \times 32 \times 10^{-3} = 0.32 \text{ W Ans.}$$

Example 4.8. A zener voltage regulator has variable load R_L requiring load current to vary 10 mA to 85 mA, given that $V_Z = 10 \text{ V}$, $I_{Z \min} = 15 \text{ mA}$; $I_{Z \max} = 100 \text{ mA}$ and the series resistance $R_S = 40 \Omega$. Calculate the range of variation permissible and zener power dissipation.

[G.G.S.I.P. Univ. Analog Electronics, December-2002]

Solution:

$$\text{Minimum supply current, } I_{S \min} = I_{L \max} + I_{Z \min} = 85 + 15 = 100 \text{ mA}$$

$$\text{Maximum supply current, } I_{S \max} = I_{L \min} + I_{Z \max} = 10 + 100 = 110 \text{ mA}$$

$$\text{Maximum supply voltage, } V_{S \max} = I_{S \max} R_S + V_Z = 110 \times 10^{-3} \times 40 + 10 = 14.4 \text{ V}$$

$$\text{Minimum supply voltage, } V_{S \min} = I_{S \min} R_S + V_Z = 100 \times 10^{-3} \times 40 + 10 = 14 \text{ V}$$

i.e., supply voltage variation is from 14 V to 14.4 V Ans.

Example 4.9. A zener diode regulator has $V_Z = 12 \text{ V}$. The input voltage may vary from 20 to 35 V and load current from 15 to 75 mA. To hold voltage constant under all conditions, what should be the value of series resistance?

[Pb. Technical Univ. Analog Electronics, December-2002]

Solution: Maximum value of required series resistance,

$$R_{S \max} = \frac{V_{S \min} - V_Z}{I_{L \max}} = \frac{20 - 12}{75 \times 10^{-3}} = 106.7 \Omega$$

Select $R_S = 100 \Omega$, standard value Ans.

Example 4.10. Calculate the values of R_S and R_L to maintain $V_L = 12 \text{ V}$ for I_L to vary from 0 to 200 mA (refer to Fig. 4.13). Also find V_Z and $P_{\max}$.

[G.B. Technical Univ. Electronics Engineering Odd Semester, 2012-13]

Solution:

$$\text{Load voltage, } V_L = 12 \text{ V}$$

$$\text{Minimum value of load current, } I_{L \min} = 0 \text{ A}$$

$$\text{Maximum value of load current, } I_{L \max} = 200 \text{ mA}$$

$$\begin{aligned}\text{Minimum value of supply current, } I_{S \min} &= I_{L \max} + I_{Z \min} \\ &= 200 + 0 = 200 \text{ mA}\end{aligned}$$

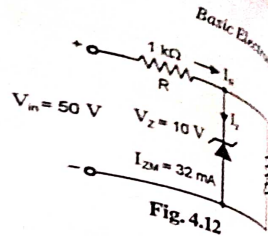


Fig. 4.12

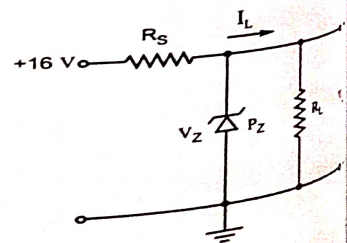


Fig. 4.13

Solution:

$$\begin{aligned}\text{Input voltage, } V_{in} &= 50 \text{ V} \\ \text{Series resistance, } R &= 1 \text{ k}\Omega \\ V_Z &= 10 \text{ V} \\ I_{Zm} &= 32 \text{ mA}\end{aligned}$$

$$\text{Supply current, } I_R = \frac{V_{in} - V_L}{R} = \frac{50 - 10}{1,000} = 40 \text{ mA}$$

Load current I_L will be maximum when zener current

$$I_Z = 0$$

$$\text{So } I_{L \max} = I_R - I_{Z \min} = 40 - 0 = 40 \text{ mA}$$

Corresponding load resistance will be minimum and

$$R_{L \min} = \frac{V_L}{I_{L \max}} = \frac{V_Z}{I_{L \max}} = \frac{10}{40 \times 10^{-3}} = 250 \Omega$$

Load current I_L will be minimum when I_Z is maximum i.e., 32 mA.

$$\text{So } I_{L \min} = I_R - I_{Z \max} = 40 - 32 = 8 \text{ mA}$$

Corresponding load resistance will be maximum and

$$R_{L \max} = \frac{V_L}{I_{L \min}} = \frac{10}{8 \times 10^{-3}} = 1.25 \text{ k}\Omega$$

Hence range of R_L : From 250 Ω to 1.25 k Ω Ans. and range of I_L : From 8 mA to 40 mA Ans.

$$\text{Maximum wattage rating of diode} = V_Z \times I_{Z \max} = 10 \times 32 \times 10^{-3} = 0.32 \text{ W Ans.}$$

Example 4.8. A zener voltage regulator has variable load R_L requiring load current to vary 10 mA to 85 mA. given that $V_Z = 10 \text{ V}$, $I_{Z \min} = 15 \text{ mA}$; $I_{Z \max} = 100 \text{ mA}$ and the series resistance $R_S = 40 \Omega$. Calculate the range of variation permissible and zener power dissipation. [G.G.S.I.P. Univ. Analog Electronics, December-2012]

Solution:

$$\text{Minimum supply current, } I_{S \min} = I_{L \max} + I_{Z \min} = 85 + 15 = 100 \text{ mA}$$

$$\text{Maximum supply current, } I_{S \max} = I_{L \min} + I_{Z \max} = 10 + 100 = 110 \text{ mA}$$

$$\text{Maximum supply voltage, } V_{S \max} = I_{S \max} R_S + V_Z = 110 \times 10^{-3} \times 40 + 10 = 14.4 \text{ V}$$

$$\text{Minimum supply voltage, } V_{S \min} = I_{S \min} R_S + V_Z = 100 \times 10^{-3} \times 40 + 10 = 14 \text{ V}$$

i.e., supply voltage variation is from 14 V to 14.4 V Ans.

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$$R_{S \max} = \frac{V_{S \min} - V_Z}{I_{L \max}} = \frac{20 - 12}{75 \times 10^{-3}} = 106.7 \Omega$$

Select $R_S = 100 \Omega$, standard value Ans.

Example 4.10. Calculate the values of R_S and R_L to maintain $V_L = 12 \text{ V}$ for I_L to vary from 0 to 200 mA (refer to Fig. 4.13). Also find V_Z and $P_{\max}$.

[G.B. Technical Univ. Electronics Engineering Odd Semester, 2012-13]

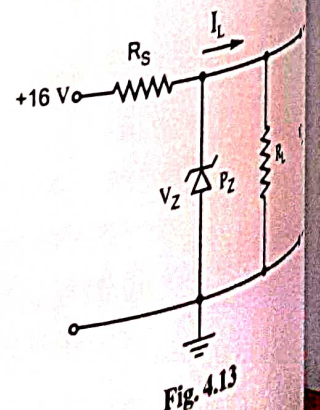
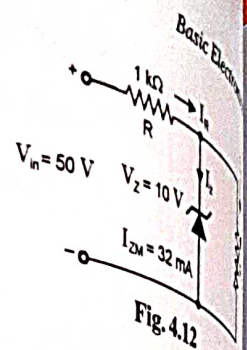
Solution:

$$\text{Load voltage, } V_L = 12 \text{ V}$$

$$\text{Minimum value of load current, } I_{L \min} = 0 \text{ A}$$

$$\text{Maximum value of load current, } I_{L \max} = 200 \text{ mA}$$

$$\begin{aligned}\text{Minimum value of supply current, } I_{S \min} &= I_{L \max} + I_{Z \min} \\ &= 200 + 0 = 200 \text{ mA}\end{aligned}$$



$$\text{Series resistance, } R_S = \frac{V_S - V_L}{I_{S\min}} = \frac{16 - 12}{200 \times 10^{-3}} = 20 \, \Omega \text{ Ans.}$$

$$\text{Minimum value of load resistance, } R_{L\min} = \frac{V_L}{I_{L\max}} = \frac{12}{200 \text{ mA}} = 60 \, \Omega$$

$$\text{Maximum value of load resistance, } R_{L\max} = \frac{V}{I_{L\min}} = \frac{12}{0} = \infty \text{ i.e. open-circuit}$$

Thus, R_L varies from $60 \, \Omega$ to ∞ Ans.

$$\text{Zener voltage, } V_Z = V_L = 12 \text{ V Ans.}$$

$$\text{Maximum value of zener current, } I_{Z\max} = I_{L\max} - I_{L\min} = 200 - 0 = 200 \text{ mA}$$

$$\text{Maximum power dissipation through zener diode, } P_{Z\max} = V_Z \times I_{Z\max} = 12 \times 200 \times 10^{-3} = 2.4 \text{ W Ans.}$$